Experiment Title: Searching Files and Content in Linux Terminal

Topic: Locating Files with `find` and Searching Text Content with `grep`

1. Objective

To practice file and text searching commands in Linux:

- Find files by name and type using **`find`**.
- Search text inside files using **`grep`**.
- Use **`grep`** flags: case-insensitive (**`-i`**), line numbers (**`-n`**), invert (**`-v`**).
- Search folders recursively using **`grep -r`** and **`grep -rn`**.

2. Theoretical Background

Linux provides two main search tools: **`find`** locates files on disk, while **`grep`** searches text inside files.

2.1 Commands Practiced in This Lab

Command	Syntax	Simple Meaning / Purpose	Example
`find`	`find <dir> -name "<pattern>"`	Find files matching name pattern.	`find . -name "*.log"`
`find -iname`	`find <dir> -iname "<pattern>"`	Case-insensitive name search.	`find . -iname "notes*"`
`find -type`	`find <dir> -type <f|d>`	Filter by file (`f`) or directory (`d`).	`find . -type d`
`grep`	`grep "<text>" <file>`	Search text inside file.	`grep "ERROR" server.log`
`grep -i`	`grep -i "<text>" <file>`	Case-insensitive text search.	`grep -i "error" server.log`
`grep -n`	`grep -n "<text>" <file>`	Show matching line numbers.	`grep -n "fail" server.log`
`grep -v`	`grep -v "<text>" <file>`	Invert search (lines without text).	`grep -v "DEBUG" app.log`
`grep -r`	`grep -r "<text>" <dir>`	Search recursively in all folders.	`grep -r "database" .`
`grep -rn`	`grep -rn "<text>" <dir>`	Recursive search with line numbers.	`grep -rn "TODO" .`

2.2 Key Differences: Find vs. Grep

Feature	`find`	`grep`
Primary Goal	Locate files and folders on disk.	Locate lines of text inside files.
Search Target	File names, types, sizes.	Text words and patterns.
Analogy	Finding a book in a library.	Finding a sentence inside a book.

3. Equipment / Software Requirements

Item	Details	Item	Details
Operating System	Linux (Ubuntu 24.04 LTS)	**Host Machine**	`firefly`
User Account	`ratul`	**Command Shell**	GNU Bash 5.2
Terminal Emulator	Linux Terminal (Kitty / Ghostty)	**Search Tools**	GNU Findutils (`find`), GNU Grep (`grep`)

4. Procedure / Steps

1. Navigate to practice directory **lab5_search**.
2. Find .log files using **find . -name "*.log"**.
3. Find files case-insensitively using **find . -iname "readme*"**.
4. Filter by type using **find . -type d** and **find . -type f**.
5. Search for "ERROR" in log file using **grep**.
6. Search case-insensitively (**grep -i**) and with line numbers (**grep -n**).
7. Invert search using **grep -v** to exclude lines.
8. Search recursively across directories using **grep -r** and **grep -rn**.

5. Result / Output Demonstration

Commands run on Ubuntu under user **ratul**:

5.1 Session 1: Finding Files by Name, Extension, and Type with find

ratul@firefly: ~/lab5_search — bash

```
ratul@firefly:~/lab5_search$ find . -name "*.log"
./logs/error.log
./logs/server.log
ratul@firefly:~/lab5_search$ find . -iname "readme*"
./docs/README.md
ratul@firefly:~/lab5_search$ find . -type d
.
./docs
./src
./logs
ratul@firefly:~/lab5_search$ find . -type f -name "*.py"
./src/main.py
./src/utils.py
ratul@firefly:~/lab5_search$ _
```

Explanation of Session 1 Results:

- **find . -name "*.log"**: Found all .log files in current directory and subdirectories.
- **find . -iname "readme*"**: Matched README.md regardless of case.
- **find . -type d**: Listed only directories.
- **find . -type f -name "*.py"**: Listed only regular Python files.

5.2 Session 2: Searching File Content with `grep`, `grep -r`, and Flags

ratul@firefly: ~/lab5_search — bash

```
ratul@firefly:~/lab5_search$ grep "ERROR" logs/server.log
[09:15:20] ERROR Database connection pool exhausted on port 5432
[09:22:45] ERROR Worker node 3 failed health check

ratul@firefly:~/lab5_search$ grep -i "warning" logs/server.log
[09:10:02] WARN  High memory threshold exceeded: 82%
[09:12:30] Warning: Slow query response time (420ms)

ratul@firefly:~/lab5_search$ grep -n "failed" logs/server.log
24:[09:22:45] ERROR Worker node 3 failed health check

ratul@firefly:~/lab5_search$ grep -v "INFO" logs/server.log | head -n 2
[09:10:02] WARN High memory threshold exceeded: 82%
[09:15:20] ERROR Database connection pool exhausted on port 5432

ratul@firefly:~/lab5_search$ grep -r "database" .
src/main.py:DB_HOST = "database.internal.net"
docs/notes.txt:Remember to run database migrations before deploy

ratul@firefly:~/lab5_search$ grep -rn "TODO" .
src/utils.py:14:# TODO: add token authentication
docs/notes.txt:3:- TODO: verify backup cron job

ratul@firefly:~/lab5_search$ _
```

Explanation of Session 2 Results:

- `grep "ERROR"`: Printed lines containing exact word `ERROR`.
- `grep -i`: Matched both `WARN` and `Warning` (case-insensitive).
- `grep -n`: Displayed matching line number (`24:`).
- `grep -v`: Excluded lines containing `INFO`.
- `grep -r`: Recursively searched all files in folders for `"database"`.
- `grep -rn`: Recursively searched and displayed file names and line numbers for `"TODO"`.

6. Conclusion

Key takeaways:

1. **`find`** locates files and directories by name, extension, and type.
2. **`grep`** searches text inside files using flags like **`-i`**, **`-n`**, and **`-v`**.
3. Recursive search (**`grep -r`**, **`grep -rn`**) quickly finds text across entire directory trees.